CTE 2059 - Computer Graphic
Lecture 2 – Materials & Lights

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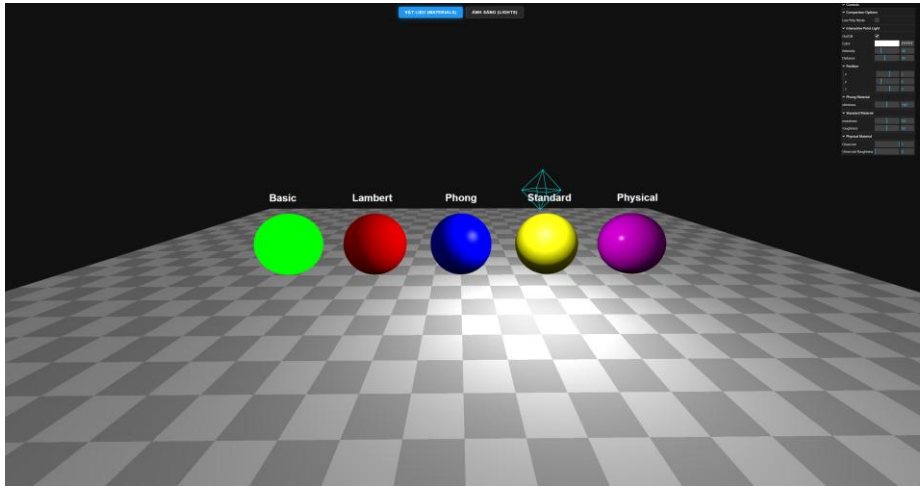
Outline

- **Getting Started**
  - Reviewing Assignment 1
  - Installation
- **Materials**
  - How to set material properties?
  - Commonly used types of material
  - Other types of material
- **Lights**
  - Commonly used types of light

Getting Started

Installation

- **Open VSCode and create a new folder**
- **Download the source code for Lecture 2 from the UET Portal**
- **Open a terminal and type the following command:**
  - npm install lil-gui
- **Run the source code. You should see a webpage with two tabs:**
  - Materials
  - Lights



Materials

How to set material properties

Materials define how object will appear in the scene.
There are 2 ways to set most material properties.

At creation time: Best for static or default values that won't change.

```
const material = new THREE.MeshPhongMaterial({
  color: 0xFF0000,    // red (can also use a CSS color string here)
  flatShading: true,
});
```

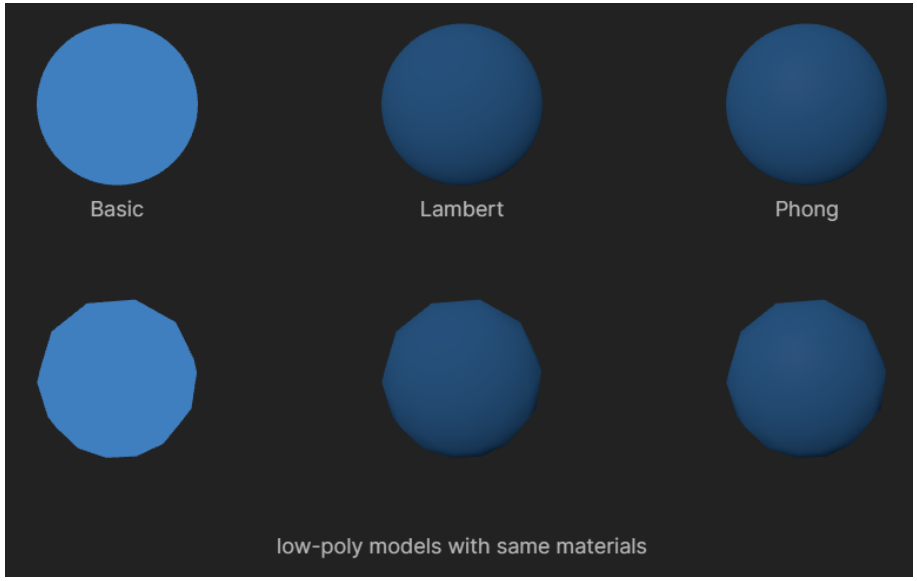
After creation time: Ideal for dynamic updates (e.g., animations, runtime conditions).

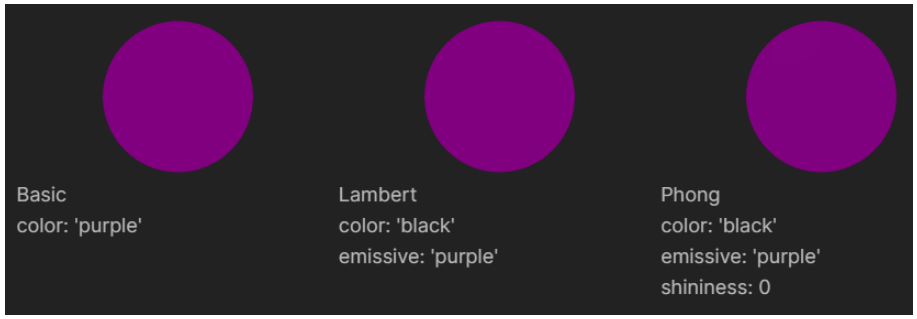
```
const material = new THREE.MeshPhongMaterial();
material.color.setHSL(0, 1, .5);  // red
material.flatShading = true;
```

Commonly used types of material

There are 3 commonly used non-PBR materials: Basic, Lambert, and Phong

- **Basic Material** is not affected by lights
- **Lambert Material** computes lighting only at the vertices
- **Phong Material** computes lighting at every pixel and support specular highlights.

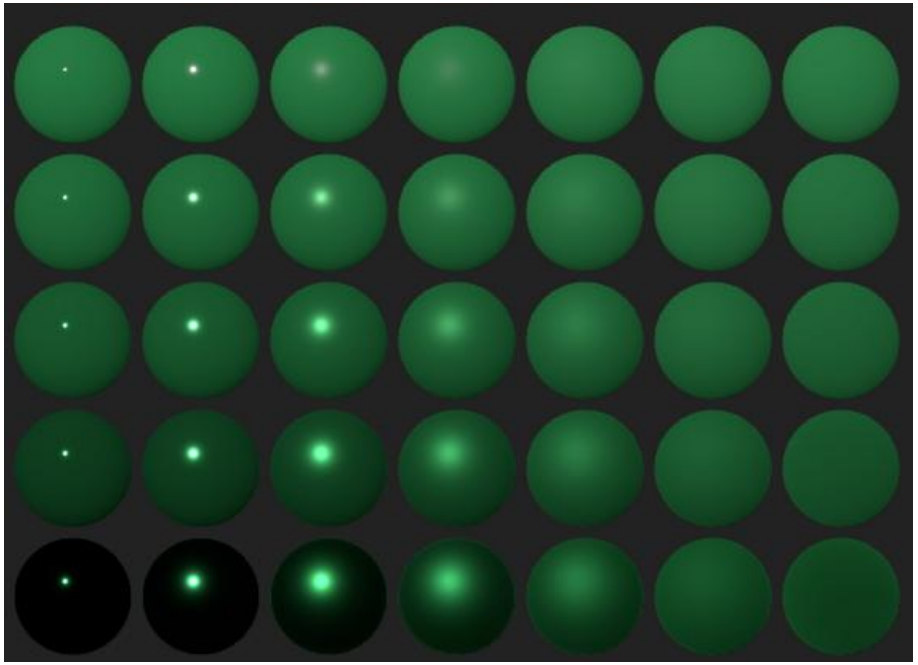


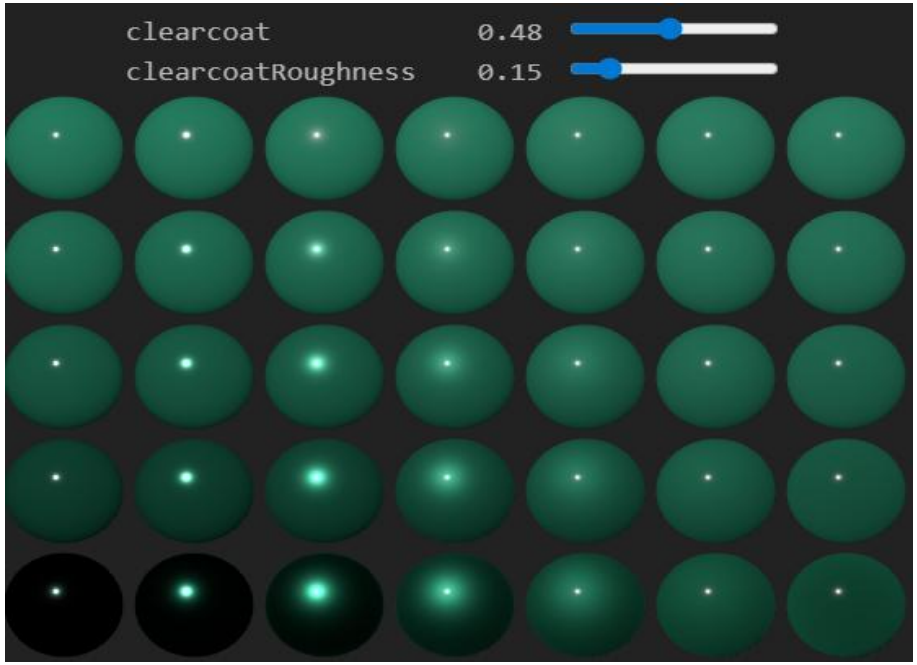


Commonly used types of material

There are 2 commonly used PBR materials: Standard, Physical

- **Standard Material** has 2 settings **roughness** and **metalness**
- **Physical Material** is same as **Standard Material**, but it adds a clearcoat gloss layer.





Commonly used types of material

- The various standard materials progress from fastest to slowest

  Basic ➞ Lambert ➞ Phong ➞ Standard ➞ Physical

- The slower materials can make more realistic looking scenes, but you might need to design your code to use the faster materials on low powered machines.

Other types of material

There are 3 materials that have special uses

- **Shadow Material** is used to get the data created from shadows.
- **Normal Material** will show the *normals* of geometry. *Normals* are the direction a particular triangle or pixel faces.
- **Shader Material** is for making custom materials using the three.js shader system.

Lights

Commonly used types of lights

There are 5 commonly used types of light

- **Ambient Light:**
  - Has no direction, lighting is 100% even
  - The AmbientLight effectively just multiplies the material's color by the light's color times the intensity.
- **Hemisphere Light:**
  - A HemisphereLight takes a sky color and a ground color and just multiplies the material's color between those 2 colors.
- **Directional Light:**
  - Often used to represent the sun.
  - We had to add the light and the light.target to the scene. A Directional Light will shine in the direction of its target.
- **Point Light:**
  - A PointLight is a light that sits at a point and shoots light in all directions from that point.
- **Spot Light:**
  - Spotlights are effectively a point light with a cone attached where the light only shines inside the cone.